

FUNDAMENTALS OF STRENGTH OF MATERIALS
PROF. D.S. PAOLINO – G. BATTIATO – C. BOURSIER
SIMULATION OF TEST 1

Surname:	Name:
ID (matricola):	

Write here the First number $P = \dots\dots$ and the Last number: $Q = \dots\dots$
of your *ID* (“matricola” on Your University card) to be used to solve the exercises
(example: $ID = 667890$: $P = 6$ and $Q = 0$)

Rules:

1. Exam time is about **2 hours**.
2. **Write very clearly the solutions** of the proposed exercises: if any lack of interpretation occurs, score will be decreased.
3. Written test is sufficient only if score is at least **18/30** and **only in this case the student can attend the second test**.
4. Student has only **15 minutes to leave this exam, after this time he/she must remain until the end and he/she must deliver all the material of the test**.
5. Solution can be only **written by means of pens with undeletable ink (no pencils) and never in red**.
6. Student **can use**: textbooks, lecture notes, didactic material of tutorials.
7. Student **cannot use**: **PORTABLE, LAPTOP, PROGRAMMABLES, IPHONE, IPOD, IPAD...** or any other electronic device that can be connected to other systems or to the network.
8. In case the Student is found working on those devices switched on, **he/she will be expelled**.
9. **Communication among students is forbidden** and in case anybody is found communicating, **he/she will be expelled**.
10. To attend the exam, each student has to show a **valid Identity Document** with a **clear picture** which could definitely certify the correspondence between the candidate and the document. In absence of this evidence the candidate will be rejected and the attendance to test shall be forbidden.
11. Each student is asked **to book the exam** to demonstrate that his/her attendance is allowed and regular. In case some do not book the exam, they will be accepted, but **if attendance will be found irregular, score of the exam will be ANNULLED** and they will repeat the exam, without any exception.
12. In case **the student attends both the written and oral tests and the attendance is then found irregular** by the Administration of Politecnico, he/she **will lose the related score** and will repeat the whole exam.

During the written test, NO QUESTIONS are allowed, being all the relevant inputs written in this document
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EX #1 (16 points)

The truss depicted in Fig.1 is made of **seven rods**:

- Rod 1: A-B;
- Rod 2: A-D;
- Rod 3: B-D;
- Rod 4: B-E;
- Rod 5: B-C;
- Rod 6: C-E;
- Rod 7: D-E.

The structure is constrained to the ground in **A** and **D**, and is subjected to an external load applied at hinge **C**. Each rod has a hollow circular cross-section with outer diameter $D = 40$ mm and inner diameter $d = 35$ mm. The rods are made of S500MC steel ($\sigma_Y = 500$ MPa, $\sigma_U = 680$ MPa, $A\% = 12$, $E = 210$ GPa).

Data: $L = 1$ m; $F = 30 \left(1 + \frac{P-Q}{100}\right)$ kN.

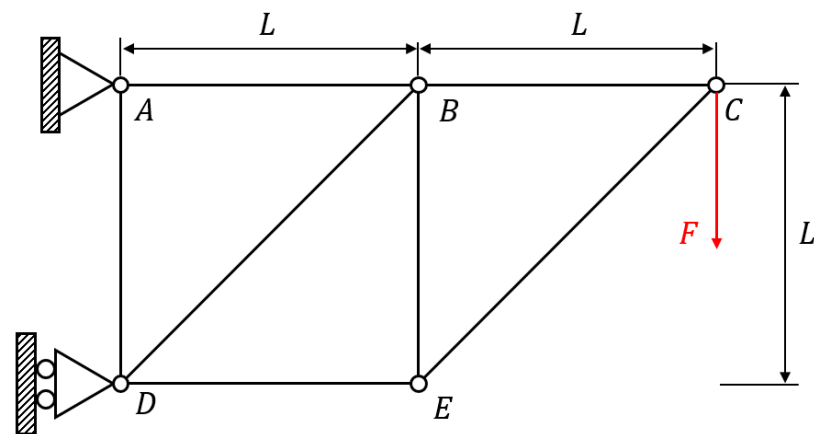


Fig. 1

Questions:

1. Compute the **degree of determinacy** h (1 point).
2. Compute the **external and the internal reactions** (4 points).
3. Compute the **internal actions** N , T_y , M_x (4 points).
4. **Plot the diagrams** of N , T_y , M_x writing their values at the relevant points (3 points).
5. Compute the **safety factors** against yielding and buckling (4 points)

EX # 2 (14 points)

Fig. 3 a) shows a cantilever beam, clamped in **A** and subjected to a uniformly distributed load $q = 10 \left(1 + \frac{P-Q}{100}\right)$ kN/m, applied between the points **A** and **B**. The beam cross-section is shown in Fig. 3 b).

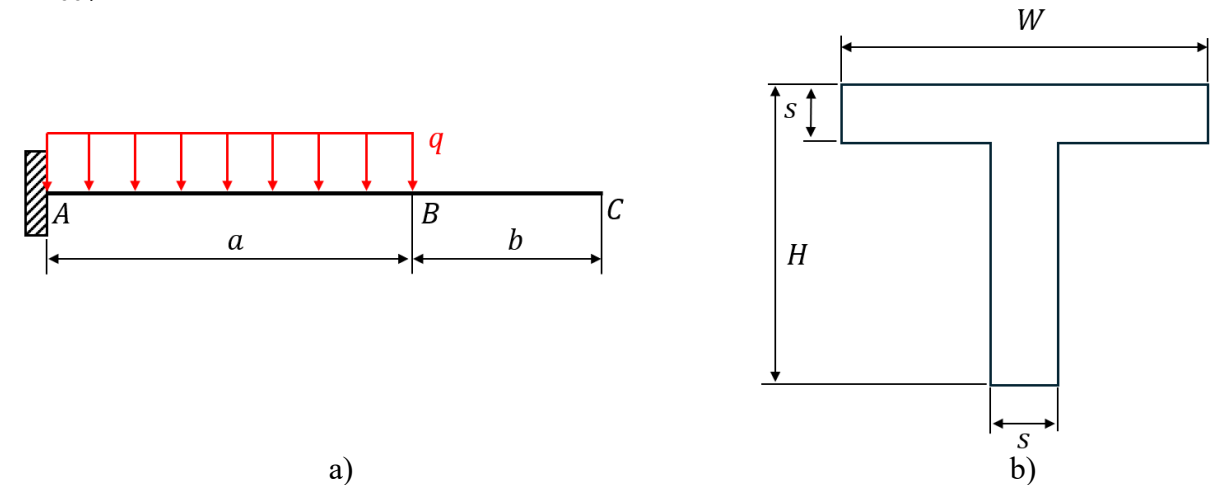


Fig. 3

The relevant beams and cross-sections dimensions are as follows:

- $a = 1$ m;
- $b = 0.5$ m;
- $W = 100$ mm;
- $H = 100$ mm;
- $s = 20$ mm.

The beam is made of steel with the following material properties: ($\sigma_Y = 400$ MPa, $\sigma_U = 450$ MPa, $A\% = 15$, $E = 210$ GPa).

Questions:

1. Compute the **degree of determinacy** h (1 point).
2. Compute the **external reactions and the internal actions** (4 points).
3. **Plot the diagrams** of N , T_y , M_x writing their values at the relevant points (3 points).
4. Compute the **safety factors** against yielding (3 points).
5. Compute the **vertical displacement** at points **B** and **C** (3 points).